SCHOOL ERP – COMPLETE MODULE & WORKFLOW OVERVIEW

1. Product Overview

The School ERP consists of two primary applications:

1. **School App:** For Principal, Vice Principal, Admin, Teaching Staff and Non-Teaching Staff.
2. **Student App:** For Students and Parents. Parent Mode is available within the same application.

A Web Portal is planned for future expansion.

The system supports multiple schools, multiple academic years, multiple staff members and multiple guardians per student.

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2. USER ROLES

A. Principal

- School registration and verification.
- School profile and branding management.
- Create Admin and Vice Principal accounts.
- Approve important school operations.
- View school-wide data, reports and analytics.
- View student and staff records.
- Manage school announcements.

B. Vice Principal

- Same operational approval authority as Principal.

- Daily school operations.
- Classroom and staff monitoring.
- Exam coordination.
- Substitute teacher management.
- Cannot create Admin accounts or manage school ownership and branding.

C. Admin

- Student and staff registration.
- Academic setup.
- Attendance management.
- Fees structure and allocation.
- Timetable and examination management.
- Notices, documents and reports.
- Student data correction with audit history.

D. Teaching Staff

- Class Teacher and Subject Teacher.
- Attendance.
- Homework.
- Marks entry.
- Timetable.
- Leave management.
- Student performance.
- Communication.
- Lesson planning and substitution requests.

E. Non-Teaching Staff

- Personal profile.
- Staff attendance.
- Leave management.
- Department tasks.
- Internal communication.
- Asset and inventory management.

F. Student

- Personal profile.
- Timetable.
- Attendance.
- Homework.
- Examination results.
- Notices and announcements.
- Academic information.

G. Parent

- Parent profile.
- Multiple children linking.
- Child attendance.
- Timetable.
- Homework.
- Results and report cards.
- School notices.
- Communication with the child's Class Teacher.

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3. PRINCIPAL MODULES

P1. School Overview Dashboard

Fields/data:

- School name and logo
- Total students
- Total teaching staff
- Total non-teaching staff
- Total classes and sections
- Present/absent students
- Staff attendance

- Pending approvals
- Academic performance
- Upcoming events

P2. Approval Center

- Request ID
- Request type
- Requested by
- School/class/section
- Request date
- Request details
- Approval status
- Approved/rejected by
- Comments
- Approval history

Approval types:

- Teacher addition/update
- Teacher assignment changes
- Timetable publication
- Fees allocation
- Results publication
- Staff leave
- Attendance approval

P3. Staff Performance

- Staff ID
- Name
- Designation
- Department
- Attendance
- Leave history
- Assigned classes
- Assigned subjects

- Performance information

P4. Student Directory

- Student ID
- Student name
- Admission number
- Roll number
- Class
- Section
- Parent details
- Contact number
- Academic year
- Student status

Principal has view access; student corrections are handled by Admin with audit logs.

P5. Academic Performance

- Class
- Section
- Subject
- Examination
- Total marks
- Obtained marks
- Pass percentage
- Class performance
- Subject performance

P6. School Calendar

- Event name
- Event type
- Start date
- End date
- Description

- Holiday status
- Target audience

P7. Staff Leave Overview

- Staff name
- Leave type
- From date
- To date
- Reason
- Supporting document
- Approval status

P8. School Announcements

- Title
- Description
- Attachment
- Audience
- Publish date
- Expiry date
- Status

P9. Admin Management

- Admin name
- Username
- Email
- Mobile number
- Account status
- Permissions

Multiple Admin accounts are supported.

P10. School Branding

- School name
- School logo

- Address
- Contact details
- School colors
- Academic identity

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4. VICE PRINCIPAL MODULES

V1. Approval Dashboard

- Pending requests
- Request details
- Approval/rejection
- Comments
- Approval history

V2. Daily Operations

- Student attendance overview
- Staff attendance
- Leave status
- Daily activities
- School operational updates

V3. Classroom Monitoring

- Class
- Section
- Assigned teacher
- Subject
- Attendance summary
- Classroom activity

V4. Exam Coordination

- Examination name

- Class
- Subject
- Exam date
- Schedule
- Teacher assignment
- Exam status

V5. Discipline Management

- Student ID
- Student name
- Class/section
- Incident type
- Incident date
- Description
- Action taken
- Remarks

V6. Substitute Teacher Management

- Original teacher
- Class/section
- Subject
- Date
- Substitute teacher
- Assignment status

V7. Staff Coordination

- Staff name
- Department
- Designation
- Assigned work
- Task status
- Remarks

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5. ADMIN MODULES

A1. School Academic Setup

- Academic year
- Class
- Section
- Subject
- Class capacity
- Class teacher
- Subject assignment
- School working days
- Academic session status

Academic year changes require Principal approval.

A2. Admission Management

- Admission number
- Student name
- DOB
- Gender
- Parent names
- Parent mobile
- Address
- Previous school
- Admission date
- Class and section
- Admission status
- Documents

A3. Fees Management

- Academic year
- Class
- Fee category

- Fee type
- General/OBC/SC/ST/EWS categories
- Fee amount
- Due date
- Concession
- Allocation status
- Approval status

Online payment gateway and payment collection are not confirmed in V1.

A4. Approval Management

- Request type
- Requester
- Request details
- Supporting documents
- Approver
- Status
- Comments
- Request history

A5. Attendance Management

Student attendance:

- Student ID
- Class
- Section
- Date
- Attendance status
- Marked by
- Remarks

Staff attendance:

- Staff ID
- Date

- Check-in
- Check-out
- Attendance status
- Attendance source (manual/biometric)
- Approval status

A6. Timetable Generator

- Academic year
- Class
- Section
- Subject
- Teacher
- Day
- Period
- Start time
- End time
- Room
- Timetable status

Admin creates timetable → sends approval request → Principal/VP approves and publishes.

A7. Examination & Result Management

- Exam name
- Exam type
- Academic year
- Class
- Section
- Subject
- Maximum marks
- Passing marks
- Obtained marks
- Grade
- Remarks

- Result status

Teacher enters marks → authorized approval → result publication.

A8. Notice & Circular Management

- Notice title
- Description
- Category
- Attachment
- Target audience
- Publish date
- Expiry date
- Status

A9. Transfer & Promotion

- Student ID
- Current academic year
- Current class
- Current section
- New academic year
- New class
- New section
- Transfer/promotion date
- Reason
- Status

A10. Reports & Analytics

- Student reports
- Staff reports
- Attendance reports
- Admission reports
- Fee allocation reports
- Examination reports
- Academic performance

- Class-wise reports
- Export options

A11. Document Management

- Document name
- Document type
- Student/staff association
- Academic year
- File
- Upload date
- Uploaded by
- Access permissions

A12. Audit Logs

- Record ID
- Module
- Action
- Old value
- New value
- Modified by
- Date/time
- Reason
- Relevant school ID

Student corrections are performed by Admin and recorded in audit logs.

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6. TEACHING STAFF MODULES

T1. Personal Timetable

- Teacher ID
- Class

- Section
- Subject
- Day
- Period
- Time

T2. Student Attendance

- Student ID
- Class
- Section
- Subject/period
- Date
- Present/Absent/Late
- Remarks

Class Teacher and Subject Teacher have separate attendance responsibilities.

T3. Homework Management

- Homework title
- Description
- Class
- Section
- Subject
- Attachment
- Assigned date
- Submission deadline
- Submission status

T4. Marks Entry

- Examination
- Class
- Section
- Subject

- Student
- Maximum marks
- Obtained marks
- Remarks

T5. Leave Management

- Leave type
- Start date
- End date
- Reason
- Supporting document
- Approval status

T6. Class Teacher Dashboard

- Assigned class
- Section
- Student list
- Attendance summary
- Student details
- Class announcements
- Class performance

T7. Student Performance

- Student
- Subject
- Examination
- Marks
- Attendance
- Performance history
- Remarks

T8. Communication

- Recipient

- Message
- Attachment
- Date/time
- Delivery/read status

Parent communication is restricted to the child's Class Teacher.

T9. Substitution Requests

- Teacher
- Date
- Period
- Class
- Subject
- Reason
- Request status

T10. Lesson Planning

- Class
- Section
- Subject
- Chapter
- Topic
- Learning objectives
- Planned date
- Completion status

Teacher assignments can change with approval and must retain historical records.

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7. NON-TEACHING STAFF MODULES

N1. Personal Profile

- Employee ID
- Full name
- Designation
- Department
- Mobile
- Email
- Address
- Joining date

N2. Attendance

- Employee ID
- Date
- Check-in
- Check-out
- Attendance status
- Attendance source

N3. Leave Management

- Leave type
- Start date
- End date
- Reason
- Approval status

N4. Task Management

- Task ID
- Task title
- Description
- Assigned by
- Assigned date

- Deadline
- Priority
- Status

N5. Internal Communication

- Sender
- Recipient
- Message
- Attachment
- Date/time

N6. Asset & Inventory

- Asset ID
- Asset name
- Category
- Quantity
- Assigned department
- Assigned employee
- Purchase date
- Condition
- Status

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8. STUDENT APP MODULES

S1. Student Login

- School-issued username
- Password
- Account status
- Session information

S2. Student Profile

Mandatory:

- Full name
- Parent names
- Parent mobile
- Gender
- Roll number
- Class
- Section
- DOB

Optional:

- Student mobile
- Email
- Address
- Emergency contact
- Height
- Weight
- Blood group
- Hobbies

S3. Student Dashboard

- Student name
- School name/logo
- Class and section
- Attendance summary
- Today's timetable
- Homework
- Notices
- Upcoming exams

S4. Timetable

- Day
- Period

- Subject
- Teacher
- Start time
- End time

S5. Attendance

- Date
- Subject/class
- Attendance status
- Monthly attendance
- Attendance percentage

S6. Homework

- Subject
- Homework title
- Description
- Assigned date
- Deadline
- Attachment
- Submission status

S7. Examination & Results

- Examination
- Subject
- Maximum marks
- Obtained marks
- Grade
- Result
- Report card

S8. Notices & Announcements

- Notice title
- Description

- Attachment
- Date
- Holiday information

S9. Student Documents

- Document name
- Document type
- Upload/download permission
- Academic year

Document access permissions require final confirmation.

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9. PARENT APP MODULES

PA1. Parent Registration & Login

- Registered mobile number
- OTP verification
- Parent name
- Email
- Account status

PA2. Child Linking

- School code
- Student admission number
- Single-use link code
- Parent/guardian verification
- Linking status

Username alone must never grant access to a child's records.

PA3. Multiple Children

- Child name
- School name/logo
- Class
- Section
- Admission number
- Child switcher

One parent can access multiple children, including children from different schools.

PA4. Child Dashboard

- Student information
- Attendance
- Timetable
- Homework
- Results
- Notices
- Messages

PA5. Attendance Monitoring

- Date
- Attendance status
- Monthly attendance
- Attendance percentage

PA6. Homework Monitoring

- Subject
- Homework
- Deadline
- Submission status

PA7. Results & Report Cards

- Examination
- Subject

- Marks
- Grade
- Result
- Report card

PA8. Communication

- Class Teacher
- Message
- Attachment
- Date/time
- Message status

PA9. Notices & Holidays

- Notice
- Holiday
- Event
- Academic calendar

Multiple guardians per student are supported. Guardian verification, unlinking and exceptional linking policies need final confirmation.

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10. COMMON SYSTEM MODULES

These are shared backend capabilities, not separate user-facing apps.

C1. Authentication & Sessions

- User ID
- Phone/email
- Password hash
- Login status
- Access token
- Refresh token hash

- Session ID
- Device information
- Expiry
- Revocation

C2. User & Role Management

- User ID
- Role
- School membership
- Permission
- Account status
- Created by

C3. Multi-Tenant School Management

- School ID
- School name
- School code
- Address
- Status
- Membership
- Academic year

Every school-owned record must be isolated by school ID.

C4. Approval Engine

- Request ID
- Request type
- Requested by
- Approver
- Status
- Comments
- Approval history

C5. Notification System

- Notification ID
- User ID
- Notification type
- Title
- Message
- Related record
- Delivery channel
- Read status
- Created date

Channels include in-app and push by default; email and SMS policies need finalization.

C6. File Storage

- File ID
- School ID
- File name
- File type
- Storage key
- File size
- Uploaded by
- Access permissions

C7. Background Jobs

- Job ID
- Job type
- Payload
- Status
- Retry count
- Created date
- Completed date

C8. Audit & Security

- User activity

- Sensitive record changes
- Login history
- Permission checks
- Administrative actions
- Security events

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11. IMPORTANT END-TO-END WORKFLOWS

Workflow 1: Principal Onboarding

Principal selects role → verifies mobile → enters profile details → selects or claims school → submits verification → Platform verification → school approval → Principal dashboard.

Workflow 2: School Setup

Principal creates Admin/VP accounts → Admin completes academic setup → creates academic year, classes, sections and subjects → Principal approves academic year changes.

Workflow 3: Teacher Onboarding

Admin enters teacher information → submits request → Principal/VP approves → account credentials generated → teacher logs in → completes profile → receives assignments.

Workflow 4: Student Admission

Admin registers student → enters mandatory information → assigns class/section → creates student account → student logs in → completes optional profile information.

Workflow 5: Parent Linking

Parent verifies registered mobile → enters school code, admission number and valid link code → system verifies guardian relationship → child becomes accessible → parent views child dashboard.

Workflow 6: Attendance

Teacher logs in → selects class/section or subject → marks attendance → validates and submits → attendance saved → reports and authorized notifications updated.

Staff attendance supports manual entry and future biometric integration.

Workflow 7: Timetable

Admin creates timetable → validates teacher/class/period conflicts → submits for approval → Principal/VP approves → timetable published → staff, students and parents receive updates.

Workflow 8: Examination & Results

Admin creates examination → configures subjects and marks → teachers enter marks → validation → approval → publication → student and parent access report cards.

Workflow 9: Fees Allocation

Admin creates fee structure → assigns fee categories and amounts → submits allocation → Principal/VP approval → approved allocation becomes visible.

Workflow 10: Leave

Staff applies for leave → authorized approver reviews → approves/rejects → attendance and substitution workflows can be updated.

Workflow 11: Student Correction

Admin opens student record → edits information → records reason → saves changes → audit log stores old and new values, editor and timestamp.

Workflow 12: Transfer & Promotion

Admin selects student → chooses new academic year/class/section → validates eligibility → processes promotion or transfer → preserves previous enrollment history.

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12. ACCESS & APPROVAL RULES

Operation	Responsible role	Approval
Create Admin	Principal	Principal
Create Vice Principal	Principal	Principal
Add teacher	Admin	Principal/VP
Update teacher	Admin	Principal/VP
Assign/change teacher	Admin	Principal/VP
Add student	Admin	No approval

Correct student data	Admin	Audit log required
Create timetable	Admin	Principal/VP
Publish timetable	Principal/VP	Authorized publisher
Enter marks	Teacher	Result approval workflow
Allocate fees	Admin	Principal/VP
Change academic year	Admin	Principal
Staff leave	Staff	Authorized approver

Approval permissions must be enforced on the backend, not just hidden in the user interface.

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13. FUTURE EXPANSION

Potential future features, not automatically part of V1:

- Online fee payments and receipts
- Advanced biometric attendance integrations
- Advanced reports and analytics
- Parent multiple-guardian management improvements
- Web Portal
- Advanced timetable optimization
- More communication and messaging capabilities
- Advanced document workflows

14. SCOPE BOUNDARY

The product scope covers school administration, academic operations, staff, students, parents, attendance, examinations, fees allocation, communication, documents, reports and security.

IT support, technical support tickets, device management and technical monitoring are excluded from this product scope.

Features marked as future, proposed or pending confirmation must not be treated as committed V1 functionality.

End of Product Module & Workflow Overview.